

Wigner-time structure of one-loop graviton dissipation in de Sitter space: certified absorptive coefficients through sixth order in H and a no-go for stationary dressings

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Publication note: this document is the frozen scientific manuscript of the GRUT research program's terminal calculation campaign, prepared for open deposit. The scientific content was frozen before this preparation; the present version differs from the frozen text only in this front matter, the completed bibliography, and typesetting. All verification is in-house; no external review has yet occurred (§7, item 9).

Author/Scope note. What is calculated here: the Wigner-time-resolved absorptive part of the one-loop pure-graviton self-energy on de Sitter background, through sixth order in H , under a fully declared contract, with representation tests and their certificates. What is inherited from prior work: the underlying self-energy object, diagram class, and Schwinger-Keldysh retarded conversion (§2, §6.5). What is not claimed: any observational prediction, any resolution of the de Sitter secular/gauge debate, any result beyond the declared contract and its stated domain (§6.4, §7).

Abstract

We compute the Wigner-time-resolved absorptive (dissipative) part of the one-loop pure-graviton self-energy on a de Sitter background, within a fully declared contract (transverse-traceless internal legs, adiabatically dressed mode functions, zero external spatial momentum, dimensional continuation), through sixth order in the expansion rate H . With $x = H \cdot u_b$ (u_b the Wigner time) and $y = H/\omega$, the certified result is

$$\text{Im } \Sigma_R(\omega > 0) = (\omega^4/1280\pi) \cdot P(x, y), \quad P = -3 - (104/3)y^2 + 24x^3 - (472/3)xy^2 - 18x^4 + 220x^2y^2 - 127y^4 - 48x^6 + (3448/3)x^4y^2 - 3312x^2y^4 - (1280/3)y^6,$$

valid for $\omega \gtrsim 3.4H$, with the H^5 sector quarantined (extracted in its delta-class only; see §7). The H^1 sector vanishes exactly; H^3 supplies the first u_b -odd (time-asymmetric) dissipative term. We establish, by a parameter-free coefficient certificate, that no representation of the form $F(Hu_b) \cdot (\tilde{\omega}^4/1280\pi) \cdot T(H/\tilde{\omega})$ with $\tilde{\omega} = \omega \cdot g(Hu_b)$ — encompassing all time-dependent amplitude dressings, frequency rescalings, and their combinations, corresponding at leading adiabatic order to the action of local field rescalings and time reparametrizations — reproduces the certified coefficients. Under the frozen retarded convention, and assuming the retarded kernel is real at fixed u_b , the certified non-analytic absorptive terms reconstruct for $\Delta > 0$ as power-law memory tails; the u_b -dependent contributions occur in the faster-decaying tails (Δ^{-5} , Δ^{-3} , Δ^{-1}), while the deepest certified infrared contribution (ω^{-2}) is u_b -independent at this order. Local counterterms cannot alter the $\Delta > 0$ content of this certified non-analytic absorptive sector. No observable is claimed; the calculation is a result of standard quantum field theory on de Sitter space.

1. Introduction

How does the dissipative response of the gravitational vacuum depend on cosmic time? On a de Sitter background the question is sharp: the background has no global timelike Killing vector adapted to the spatially flat slicing, so a two-point response kernel $\Sigma_R(t, t')$ need not be a function of $t - t'$ alone. Whether — and in precisely what structural way — the one-loop graviton self-energy departs from stationarity is a well-posed, computable question, and this paper answers it at the level of the absorptive (imaginary, dissipative) part, through sixth order in H , within a fully declared calculational contract.

The result is presented in the order of its logical construction: the calculation (§2); the certified absorptive structure (§3); the exact Fourier/distributional mapping to relative-time memory structure, with its single physical premise stated explicitly (§4); the representation test — a no-go certificate for stationary dressings (§5); physical interpretation (§6); limitations (§7); and what remains unresolved (§8). Section 9 records provenance. This ordering is deliberate: every interpretive statement in §6 is downstream of a certificate or an exact mapping in §3–§5, and the qualifications of §7 are part of the result, not caveats appended to it.

This work originated in a research program (GRUT) investigating a "responsive-vacuum" interpretation of gravitation; the program's own audit concluded that no result herein distinguishes that interpretation from standard quantum field theory on de Sitter space (§6.4, §9). The calculation and its certificates stand independently of any interpretation.

2. The calculation

The one-loop graviton self-energy on de Sitter backgrounds is an established object: computed by Tsamis and Woodard [TW96], with a flat-space limit matching Capper's gauge-fixed computation [C79, as matched in TW96]; recomputed from gravitons in cosmology, with a nine-structure-function representation, by Tan, Tsamis and Woodard [TTW21a]; and converted to a retarded Schwinger-Keldysh form and applied to gravitational radiation in [TTW21b], whose conversion (Σ_{ret} from $\Sigma_{++} + \Sigma_{+-}$) the present contract inherits. The Wigner-resolved absorptive construction and everything downstream of §3 are the present work's own (§6.5).

The object is the retarded self-energy of the transverse-traceless (TT) graviton on a de Sitter background of expansion rate H , at one loop in the pure-gravity sector, under the following declared contract (all elements frozen before computation; see §9 for the audit trail):

- **Internal legs:** the declared TT bath, $\langle hh \rangle = P^{\text{TT}} \times W$, with the Wightman kernels $W_{\pm} = (\kappa^2/q) e^{\mp i q(u-u')} [(1-Hu)(1-Hu') \pm i H^2(u-u')/q + H^2/q^2]$ — the exact product of linearly dressed adiabatic modes, terminating at $O(H^2)$ per kernel. TT uniqueness is not claimed.
- **Vertex:** the cubic de Sitter vertex reduced to H -graded bilinear C -matrices on the spatial pair basis (maximum H -degree 2).
- **External kinematics:** frequency ω at zero external spatial momentum, reached as a controlled $k \rightarrow 0$ limit (never the literal point), with an executed two-axis isotropy gate.
- **Radial sector:** dimensional continuation only, no explicit IR scale; angular averages at symbolic dimension d , with $d \rightarrow 3$ taken term-by-term (smoothness verified at each order).
- **Retarded convention (frozen, quoted):** $\Sigma_R = -i\theta(\Delta) \cdot \text{RET}$ with $\Sigma_R(\omega) = \int_0^\infty d\Delta e^{+i\omega\Delta} \Sigma_R(\Delta)$ — the upper-half-plane-analytic transform. The absorptive part is extracted from the $\omega = +2q$ cone of the loop integrand by the Sokhotski delta class, with a reality gate (i^{ncn} real) enforced per Δ -degree; for the present $\omega > 0$ absorptive delta-class extraction the opposite cone has no support (its delta class lies at $q = -\omega/2$, outside the measure domain; both cones carry independent content for the dispersive part — §8) — verified per degree, term by term.

- **Units:** $\kappa = 1$; all coefficients below are exact rationals over π .

The two-time structure is carried in Wigner variables: $u_b = (u+u')/2$ (the Wigner time) and $\Delta = u - u'$, with the transform taken in Δ at fixed u_b . The expansion is H-graded; the source object carries H-degree 8, of which orders 0–6 are extracted here (orders 7–8 were removed by a builder-level truncation prior to caching and are outside the certificate; §7).

3. The certified absorptive structure

The certified result, with $x = H \cdot u_b$ and $y = H/\omega$:

$$\text{Im } \Sigma_R(\omega > 0) = (\omega^4/1280\pi) \cdot P(x, y)$$

order	contribution to P	status
H^0	-3	complete; three independent routes agree
H^1	0	complete — vanishes exactly
H^2	$-(104/3) \cdot y^2$	complete; u_b -free
H^3	$+24x^3 - (472/3) \cdot x \cdot y^2$	complete — first u_b -odd term
H^4	$-18x^4 + 220x^2y^2 - 127y^4$	complete
H^5	$(416/3)x^3y^2 - 496xy^4$	delta-class only — quarantined , never binding
H^6	$-48x^6 + (3448/3)x^4y^2 - 3312x^2y^4 - (1280/3)y^6$	complete — first negative ω power

"Complete" means: extracted through gated machinery whose per-degree reality preconditions all hold, whose $d \rightarrow 3$ reduction is smooth piece-by-piece, and which reproduces the independently established H^4 coefficient exactly as a prerequisite gate on every run. At H^5 the reality precondition fails at one Δ -degree — the condition under which the extraction's own derivation states that principal-value content leaks into the absorptive part — so only the delta-class content is reported there and no binding statement uses it.

Structural features of the certified table:

1. **Validity domain.** The record's own derived boundary is $\varepsilon_H = (104/9)H^2/\omega^2$, with refusal at $\omega = \sqrt{(104/9)} \cdot H \approx 3.399H$ — the ratio of the H^2 to H^0 coefficients. The u_b -free tower $(-3, -104/3, -127, -1280/3)$ has successive ratios $\approx 11.6, 3.7, 3.4$; this three-term pattern is suggestive of a lower characteristic scale near $\omega \sim 1.8H$ but does not establish a convergence radius — the operative boundary remains the refusal at $3.399H$.
2. **Sign.** The $\leq H^4$ truncation is analytically negative on the stated window (exact maximum $-599127/270400$). The H^3/H^6 -inclusive expression was negative at every tested grid point, for both signs of u_b — grid verification, not an analytic proof. Accordingly, no gain/instability was observed within the tested domain.
3. **Time asymmetry.** The first u_b -odd absorptive contribution enters at H^3 (odd in u_b , even under the joint flip $(H, u_b) \rightarrow (-H, -u_b)$, as the construction's parity mechanism requires), establishing Wigner-time asymmetry of the dissipative coefficient. This should not be identified, by itself, with an arrow-of-time statement.
4. **Secular self-termination.** The u_b -dependence equals the flat response at $Hu_b = 6^{(-1/4)} \approx 0.64$ (at $y = 0$): the certified truncated response is restricted to Wigner drifts below this surface by the adopted self-termination criterion.

4. Fourier/distributional mapping to relative-time structure

Three layers, kept separate.

4.1 Mathematical mapping (prescription-independent; conditional on the §4.2 premise). For the certified spectral terms, and assuming the retarded kernel is real at fixed u_b , the frozen convention maps the odd extensions implied by the §4.2 reality premise to the corresponding $\Delta > 0$ contributions via $K(\Delta) = (2/\pi) \int_0^\infty \text{Im } \Sigma(\omega) \sin(\omega\Delta) d\omega$. This extracts the non-analytic tail associated with each certified term under the stated prescription; it does not reconstruct the complete physical kernel, whose spectrum is certified only on the stated frequency domain (§7, item 1). A local (contact) operator $K = c \cdot \delta^{(n)}(\Delta - 0^+)$ has $\Sigma = c(-i\omega)^n$ exactly: only odd n contributes to the absorptive part, and then as an *analytic odd polynomial*. The certified terms are even powers of $|\omega|$ — no certified term is a local-operator form. Their $\Delta > 0$ kernels, computed identically under two independent prescriptions (Abel regularization; analytic continuation $\Gamma(p+1) \cdot \sin(\pi(p+1)/2) / \Delta^{(p+1)}$) and confirmed by direct quadrature:

$$\omega^4 \rightarrow +48/(\pi\Delta^5) \quad \omega^2 \rightarrow -4/(\pi\Delta^3) \quad \omega^0 \rightarrow +2/(\pi\Delta)$$

— power-law memory tails, not contact terms. Contact/counterterm ambiguities are supported at $\Delta = 0$ only and cannot affect any $\Delta > 0$ statement.

4.2 Physical premise (stated, not derived). Extending the computed positive-frequency absorptive coefficients to the full odd spectral function $\text{sgn}(\omega)|\omega|^p$ — the step on which §4.1's reconstruction rests — requires the retarded kernel to be *real at fixed u_b* . This is a physical reality condition on the retarded response (standard for retarded responses of hermitian sources, and consistent with every gate of the extraction, including the per-degree reality preconditions), **not** something the cached loop integrand itself proves: the cached object's pointwise form is dressed by the assembly's derivative-operator routing and is not pointwise anti-hermitian. The move from computed coefficient to odd extension is therefore a stated premise of the tail statements, not an identity.

4.3 Regime limitation (the infrared window). For the ω^{-2} term (H^6 , u_b -free), the exact $\Delta > 0$ kernel of the IR-cut model — cutoff ω^* at the domain boundary — is $-\Delta \cdot \text{Ci}(\omega^* \Delta) + \sin(\omega \Delta) / \omega$: it grows as $\Delta[1 - \gamma_E - \ln(\omega \Delta)]$ **within the window** $\Delta \lesssim 1/\omega$ ($\omega^* = 3.399H$, i.e. $\Delta \lesssim 0.29/H$) and is bounded oscillation beyond, in the cut model. The calculation does not determine the asymptotic behavior beyond the window; "indefinitely growing secular tail" is *not* a licensed description.

5. The representation test: a no-go for stationary dressings

5.1 The tested class. Consider representations

$$\text{Im } \Sigma = F(x) \cdot (\tilde{\omega}^4/1280\pi) \cdot T(H/\tilde{\omega}), \quad \tilde{\omega} = \omega \cdot g(x),$$

with F, g analytic ($F(0) = g(0) = 1$) and T arbitrary through the matched order. The declared class contains time-dependent amplitude dressings ($g \equiv 1$), frequency rescalings ($F \equiv 1$), and their combinations; in the leading-adiabatic (Wigner-local) construction used here, it corresponds to the action of local field rescalings $h \rightarrow Z(u_b)h$ and time reparametrizations $u \rightarrow f(u)$ on a stationary law.

5.2 The certificate. Matching the class to the certified table on every complete slot through total order 6 proceeds as a chain of single linear equations: the pure- y slots fix $t = (-3, -104/3, -127, -1280/3)$; the measured $H^1 = 0$ and H^3 data pin the odd parameters $f_1 = 118/13$, $g_1 = -59/26$; the $(2,0)/(2,2)$ slots fix $f_2 = 1336/169$, $g_2 = 14733/1352$; and the x^2y^4 slot is then **parameter-free**: the class requires $-169672/169 \approx -1004.0$, while the complete H^6 datum is -3312 . The residual $390056/169 \neq 0$ refutes the entire class on complete data, independent of the quarantined H^5 sector (which enters no matched slot). Narrower certificates refute the amplitude-only and rescaling-only subclasses already at H^4 .

5.3 Robustness. Because the certified content is non-analytic (§4.1), local counterterms cannot generate or remove any certified coefficient: the no-go concerns content that is not a

renormalization convention. Beyond leading adiabatic order, the redefinition orbit acquires gradient corrections, bounded by the computed field $E = (H/\omega)|\partial_x P|/|P|$. At three of the fifty evaluated grid points (larger H/ω , the u_b -odd side) the certified obstruction exceeds the computed gradient-correction bound; at those points the no-go extends to the tested full orbit. Elsewhere only the leading-adiabatic statement is claimed.

5.4 The Wigner-local approximation, quantified. The same field E measures the error of replacing the two-time response by a local-frequency form: it reaches ≈ 0.33 on the tested inner domain ($\omega \geq 5H$, $|Hu_b| \leq 0.3$) and ≈ 1.68 at the $\omega \approx 3H$ edge, dominated by the u_b -odd H^3 term. On the tested inner domain the Wigner-local reduction carries an error reaching approximately 33%, and therefore is **not certified as a controlled reduction** under the preregistered 20% criterion. Any use of $G_R(\omega, k)$ in place of $G_R(t, t'; k)$ on this domain inherits that error.

6. Physical interpretation

6.1 The central statement. Within the frozen retarded convention, and assuming the retarded kernel is real at fixed u_b , the certified non-analytic absorptive terms reconstruct for $\Delta > 0$ as power-law memory tails. The u_b -dependent contributions occur in the faster-decaying tails, while the deepest certified infrared contribution is u_b -independent at this order. This statement concerns the certified non-analytic absorptive sector; local counterterms cannot alter its $\Delta > 0$ content. The time-domain behavior of the ω^{-2} contribution remains bounded by the stated infrared window, beyond which the calculation does not determine the asymptotic behavior.

6.2 Cosmic-time dependence and memory are not the same thing. The original intuition motivating this program fused them; the calculation separates them. Within the certified response, cosmic-time dependence and memory *depth* are anti-correlated: the explicitly time-dependent pieces ride the short-memory tails (Δ^{-5} , Δ^{-3} , Δ^{-1}), and the deepest certified memory contribution is stationary at this order. Equivalently, in frequency space: no time-dependent amplitude, no time-dependent clock, and no combination of the two turns the certified response into a stationary law (§5.2) — yet the component that most resembles genuine long-time memory carries no explicit cosmic-time dependence at this order.

6.3 What kind of result this is. Every statement above is a result of standard quantum field theory on de Sitter space, obtained within the declared contract. The calculation provides the Wigner-time-resolved absorptive coefficients at H^3 , H^4 , H^6 , including the exact H^1 vanishing and the first u_b -odd term, together with the stationarity no-go certificate, the tail/nonstationarity anti-correlation, and the quantified Wigner-reduction error; a supported novelty statement awaits a documented literature search. No claim of physics beyond standard QFT is made. The stationarity no-go is therefore a structural statement about the calculated self-energy, not an experimental discriminator.

To the best of our knowledge, following the literature search documented in the program record, the specific Wigner-resolved, H -graded absorptive coefficient hierarchy studied in this work — including the u_b -odd H^3 contribution — and the associated parameter-free stationary-dressing inconsistency certificate have not previously been reported in this form. The underlying de Sitter graviton self-energy, its infrared/secular behavior, absorptive self-energy phenomena in de Sitter, and the broader question of whether de Sitter time dependence can be removed or interpreted physically are established subjects with substantial prior literature (§6.5).

6.4 Interpretation programs. A "responsive-medium" reading of the gravitational vacuum is *consistent with* this structure — an expanding background carrying time-dependent response that no stationary dressing absorbs — but nothing here distinguishes such a reading from standard QFT on de Sitter; the originating program's own audit records that its prediction ledger is empty and no discriminating observable survived its gates. The

computed absorptive response is parametrically Planck-suppressed; no observable is claimed. This question was assessed in the literature pass and is not presently supportable from the certified calculation: every conversion route examined imports a dressed propagator / response calculation, establishment of the relevant gauge-invariant status of the off-shell coefficient, or the additional low-frequency/transport calculation required for the proposed observable routes.

6.5 Relation to prior work. The physical interpretation of secular effects in de Sitter quantum gravity has been debated, including arguments concerning gauge dependence, infrared regularity, and the definition of observables. The present calculation does not resolve that broader debate. Its stationarity result is instead a statement about the declared self-energy object and the specified class of amplitude and frequency dressings.

The closest existing result is [TTW21b]: solving retarded effective field equations in position space, single-graviton loops produce secular growth of graviton mode functions, $u_1 \rightarrow (\kappa^2 H^2 / 4\pi^2)(4/3)\ln^2 a \cdot u_0$, with breakdown at $\kappa^2 H^2 \ln^2 a \sim 1$. That is a loop-counting/secular-growth criterion; the present secular surface $Hu_b \approx 0.64$ is a kinematic boundary of the adopted one-loop H-graded truncation. The two belong to the same de Sitter infrared/secular physics family but are not competing estimates of the same quantity. On the physical status of such effects: the negative, infrared-divergent on-shell graviton self-energy goes back to [TW95]; critiques and gauge/slicing analyses include [DEZ94] and [GT08] (with reply [GT-R]); infrared-regularity programs include [TU14]; recent work argues gauge independence of graviton-loop logarithms once source/observer correlations are included [GIL24]; and soft-graviton corrections to matter couplings have been argued to be physical and covariance-forced, with time-dependent wave-function renormalization absorbing kinetic but not coupling corrections [KK14]. On the removal side, infrared/secular growth of *scalar correlators* is systematically absorbed by counterterms in dimensional regularization [HHPS24], treated in Soft de Sitter Effective Theory [SdSET], and resummed by dynamical renormalization group methods [DRG]. These transformations act on scalar correlation functions or matter couplings; none acts on the graviton absorptive self-energy studied here. Open-system treatments of the primordial *tensor* spectrum, with explicitly non-Markovian structure beyond the strict Markovian limit, appear in [OEFT22] — the framework family within which a two-time treatment of the present certified kernel would naturally sit. Absorptive and nonstationary self-energy phenomena on de Sitter are established for matter fields: dS-allowed decays [BEM10], initial-value nonstationary self-energies and decay widths with Bose enhancement at $T = H/2\pi$ [B-line]. Wigner/Kadanoff-Baym two-time methods are standard in nonequilibrium field theory, including matter transport in curved spacetime [KBE]; the literature search documented in the program record did not identify an application producing the Wigner-resolved absorptive graviton self-energy hierarchy studied here. Finally, the flat-space calculations [C79, TW96] provide a diagrammatic and normalization historical anchor; no coefficient identity is asserted between the present flat coefficient and the gauge-dependent off-shell coefficient of those calculations, because the compared projected objects differ in gauge and construction (general-gauge dependence of the graviton self-energy is itself classical literature [GG78]).

7. Limitations

The following are part of the result.

1. Validity domain: $\omega \gtrsim 3.4H$ within the declared truncated expansion (boundary derived; coefficient chart-dependent); no claim is made for $\omega < 3.4H$.
2. Wigner-time window: secular self-termination at $Hu_b \approx 0.64$ ($y = 0$); statements tested at $|Hu_b| \leq 0.5$.
3. H^5 quarantined (delta-class only; a principal-value completion is absent); H^7/H^8 not extracted (source object degree 8).

4. Re Σ_R not computed: no pole structure, no μ -anchored dispersive statement, no full G_R resummation.
5. The Wigner-local reduction is not certified as controlled (§5.4): $\sim 33\%$ error on the tested inner domain against a preregistered 20% criterion.
6. The full-orbit form of the stationarity no-go holds on its computed subdomain only; elsewhere the statement is leading-adiabatic-order exact.
7. Long- Δ statements beyond $\Delta \approx 0.29/H$ are not licensed; the ω^{-2} kernel's form beyond the infrared window is undetermined (§4.3).
8. One loop only; internal bath dynamics absent; the $\omega \rightarrow 0$ transport class is undetermined and is excluded here by prior programmatic reservation.
9. All verification is in-house; no external review has yet occurred.
10. Three auxiliary negatives in the wider program record carry a must-be-reopened flag and are not used here.
11. The contract's boundary declarations are temporal only; the status of the constant-TT mode under the equivalence relation is unresolved, and it is not quotiented by assumption.
12. An earlier internal labeling of the u_b -dependent terms as "local" was incorrect and is superseded by §4 (the correction is part of the program record).
13. The odd spectral extension rests on the stated reality premise (§4.2); the counterterm-invariance claim is scoped to the certified non-analytic absorptive sector, not the complete self-energy.

8. What remains unresolved

The dispersive side (Re Σ_R via principal-value integrals of both cones — the two cones carry independent content and neither is reconstructible from the other by conjugation); the H^5 completion; the deep-infrared completion $\omega \lesssim H$ and with it the transport class of the response (a question that additionally requires the bath's internal dynamics, beyond one loop); the H^7/H^8 sectors; the boundary/equivalence declaration for the constant-TT mode at spatial infinity; whether any consistent infrared completion exists in the smooth-window class after the exclusion of sharp completions; and the foundational questions the wider program leaves standing (outcome selection; the origin of $\hbar$; the emergent-versus-fundamental status of the framework), untouched by everything computed here.

9. Provenance and methods

The evidence base is frozen and versioned: (i) the program baseline GRUT_REALITY_BASELINE_01 (repository GRUT-RAI, branch v4, commit 5bdda88; registry sha256 dc693eb6...436a9), itself the output of an audited reconstruction with a quotation-verified record; (ii) the banked calculation chain T3-01 through T3-09B in the working repository (instruments, result files, and complete run logs, including disclosed instrument defects and their corrections), with provenance-pinned caches for every extracted object. Every number in this paper is mechanically verified against those artifacts before commit. Computations were performed symbolically (exact rational arithmetic throughout; numerical checks by independent quadrature where stated) with AI-assisted execution under a pre-registration discipline: outcome classes declared before each computation, adversarial gates required to pass before any number is read, and refuted premises disclosed rather than removed. Numerical-method failures encountered en route (two quadrature/extrapolation mis-setups) are recorded in the run logs as provenance; they are method episodes, not properties of the transforms, and the corrected calculations appear in §4.

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- [KBE] Kadanoff-Baym/Wigner methods in curved spacetime: e.g. arXiv:0807.4551 (transport from Kadanoff-Baym equations in curved spacetime); standard nonequilibrium-QFT texts.

Version note (v1.0): the scientific text of this manuscript was frozen under the program's verification discipline (three owner edit passes; a mechanical verification battery re-run at each freeze; positioning additions applied only after a documented literature pass). Nothing in it may be strengthened without a corresponding certificate; the thirteen limitations of §7 travel with any excerpt.